UNIVERSITI SAINS MALAYSIA

School of Electrical and Electronics Engineering
Universiti Sains Malaysia, Engineering Campus

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EEM 441

Control & Instrumentation Laboratory

Experiment 1

DC MOTOR / PID CONTROL — SPEED

Date : ______________________

Group No. : ______________________

Group Members : ______________________ Matrix # __________

: ______________________ Matrix # __________

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Experiment 1

DC Motor / PID Control — Speed

Objectives

1. To study the open-loop speed control of a DC motor.
2. To study closed-loop speed control using P, PD, PI, and PID controllers.

Equipment / Software Required

- Controller kit
- Cathode ray oscilloscope
- BNC connectors with leads
- Multimeter

Procedure

(a) Open-loop control of DC motor

1. Switch on the supply and measure all constant voltages; calibrate variable voltages in terms of voltage and angle (degrees).
2. Disconnect all cable wires from the hardware module (Figure 1.1).
3. Connect *Output A* of the motor plant to Channel 2 of the oscilloscope.
4. Connect *Adjust A* of the motor plant to *Input A* (speed control of the plant) and to Channel 1 of the oscilloscope (only positive voltages are considered).
5. Turn ON the power switch.
6. Slowly turn potentiometer-A clockwise (0 to $+50$) and anti-clockwise (0 to -50). Observe and describe the motor operation.
7. Disconnect *Adjust A* and connect the motor *Input A* to *Adjust B* (only positive voltages are considered).
8. Vary supply B from $30°$ to $270°$ in steps of $30°$ and study the variation in reference speed (voltage) and output speed (voltage). Plot the error curve.
9. Comment on your observations and results.

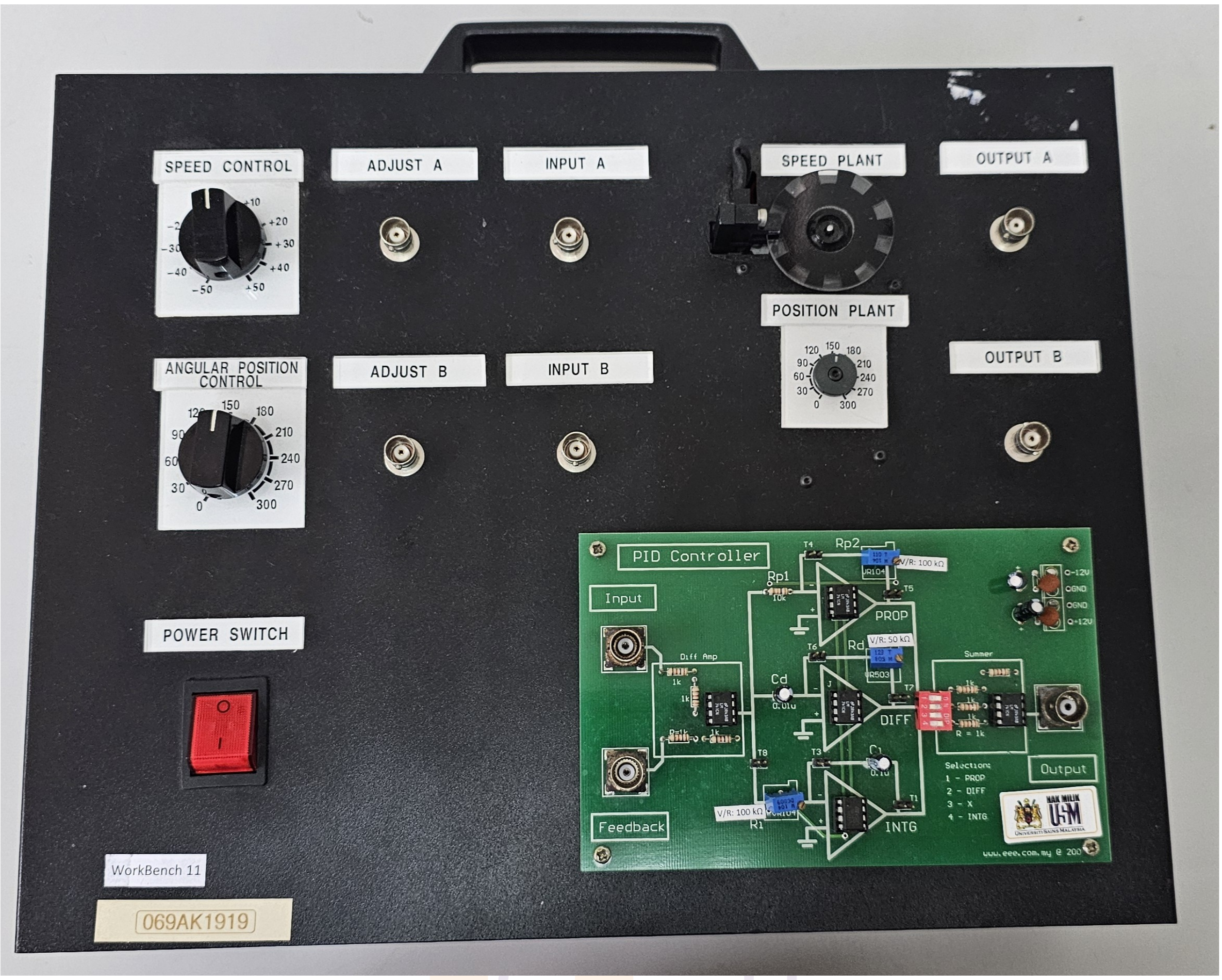

Figure 1.1: DC Motor / PID Control trainer.

Table 1.1: Open-loop speed response.

Position	V_{in} (Channel I)	V_{out} (Channel II)	Error ($V_{out} - V_{in}$)	Remarks
+270				
+240				
+210				
+180				
+150				
+120				
+90				
+60				
+30				

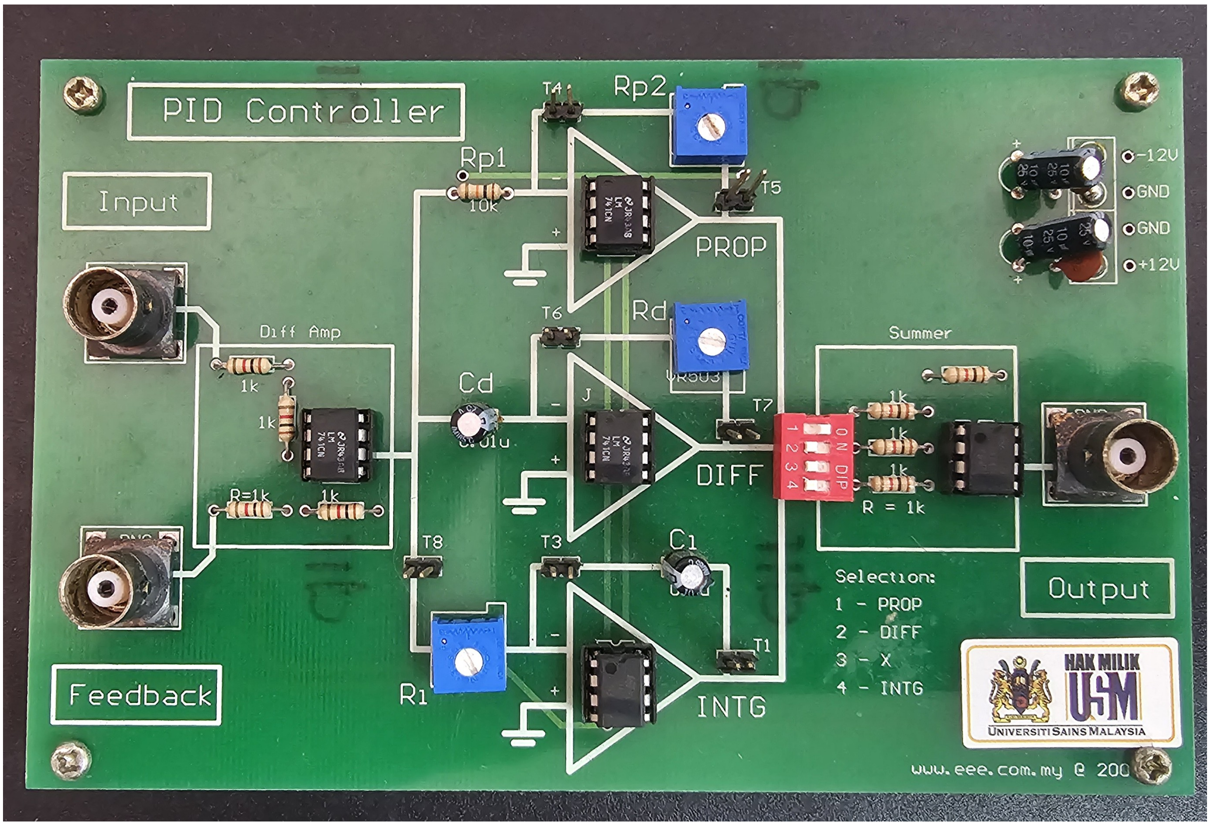

Figure 1.2: PID Control module.

(b) Closed-loop control of speed of DC motor

(i) Using P-controller with feedback through optical encoder

1. Select the P-controller by setting PROP on the controller (DIP switch 1) to ON (refer to Figure 1.2).
2. Connect *Adjust B* of the plant to the input of the PID controller.
3. Connect *Output A* of the plant to the controller feedback.
4. Connect the controller output to *Input A* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. $+180$).
6. Adjust trimmer R_{p2} (varies K_p) to obtain an optimum response (minimum steady-state error).
7. Turn potentiometer-B to vary the speed of the motor. Record the input and output voltages for various positions in tabular form.

(ii) Speed control with PI-controller

1. On the PID controller card, set trimmers R_p and R_d to minimum (counter-clockwise) while R_i is set to maximum (clockwise).
2. Select the PI-controller by setting PROP and INTG to ON.
3. Slowly vary trimmer R_i to obtain an optimum output response (minimum steady-state error). If gain is insufficient (or overshoots), adjust R_{p2}.

4. Turn potentiometer-B to vary the speed of the motor. Record the input and output voltages for various positions (Table 1.2). Repeat step 3 if necessary to fine-tune the response.
5. Compare the experimental results for the P-controller and the PI-controller.

(iii) Speed control with PD-controller

1. Switch on the supply and measure all constant voltages; calibrate variable voltages in terms of voltage and angle.
2. On the PID controller card, set trimmers R_p and R_d to minimum (counter-clockwise) while R_i is set to maximum (clockwise).
3. Select the PD-controller by setting PROP and DIFF to ON; all other switches OFF.
4. Vary supply B from $30°$ to $270°$ in steps of $30°$ and study the variation in reference speed (V_{in}) and output speed (V_o). Plot the error curve.
5. Study the effect of varying R_d on the output.

(iv) Speed control with PID-controller

1. Adjust potentiometer-B to 2 V (around $+180°$).
2. Set trimmer R_{p2} to minimum to reduce the damping ratio.
3. Set trimmer R_i to minimum. The output should start to vibrate.
4. Slightly increase R_i until the system starts oscillating between two points (pure oscillation).
5. Slightly increase R_{p2} to bring the system to a damped condition; it should oscillate for a few cycles.
6. Apply a small force to hold the motor (the oscilloscope will show the motor output drop to 0 V); observe and describe the response when the force is released.
7. Select the PID-controller by setting PROP, INTG, and DIFF to ON.
8. Slowly vary R_d to obtain the optimum response.
9. Compare its operation with the P, PI, and PD controllers under disturbance.
10. As before, take observations by varying the applied voltage.

Table 1.2: Closed-loop speed response.

Position	V_{in} (Channel I)	V_{out} (Channel II)	Error ($V_{out} - V_{in}$)	Remarks
$+270$				
$+240$				
$+210$				
$+180$				
$+150$				
$+120$				
$+90$				
$+60$				
$+30$				

Report

1. Compare the performance of P, PI, PD, and PID controllers.
2. What will happen if all types of controllers are connected in series?
3. What are the different types of encoders? Explain the working principle of an optical encoder with the help of suitable diagrams.